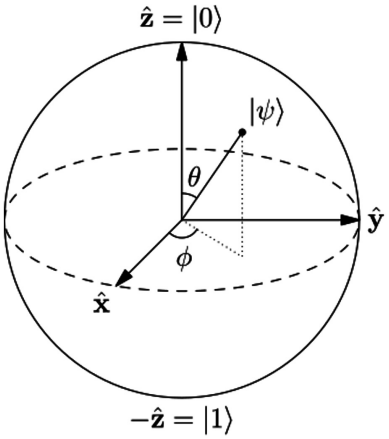
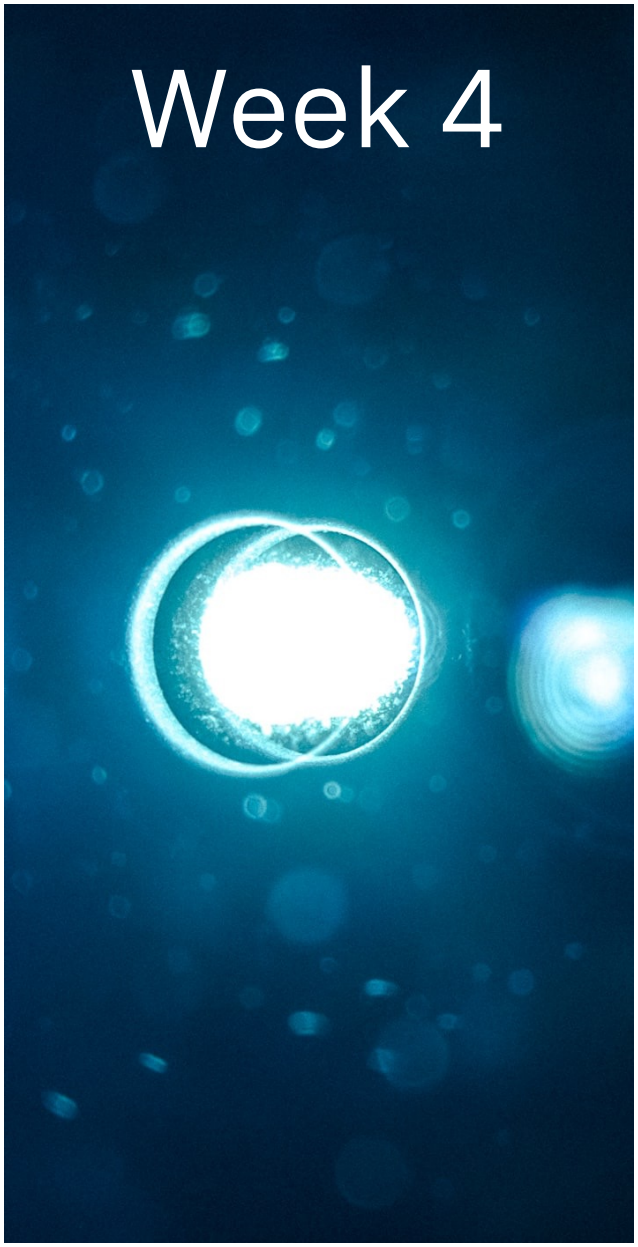
Quantum Coding with Qiskit – A Course for Beginners

3rd June 2026

Sebastian E. Grodzietzki, Dipl.-Inf.
Joey Rogers, M.Sc. Physics





Welcome & Agenda / Week 4

Week 4 – Applications Framework 🧩 / this is a **content week (no code)**

IBM Quantum Platform
- Get free Access: Intro to Open Plan
- Platform Overview (Documentation, Functions, Composer, Announcements)
- Open Plan Setup: Where to get the API Key and your free Quantum machine instance

Coding Flavors
- Quantum chemistry
- Optimization
- Quantum machine learning

Qiskit Overview
- Coding Pattern / Workflow
- Qiskit Functions
- Qiskit Adocates Program
- Qiskit Summer School

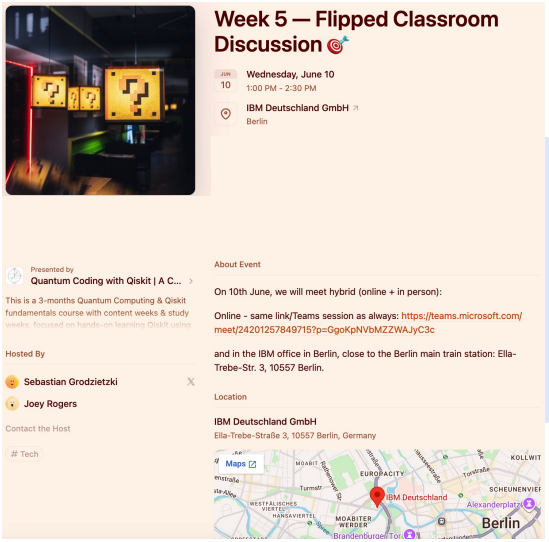
Q&A

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Before we Start...

Next week, 10th June, Flipped Classroom...



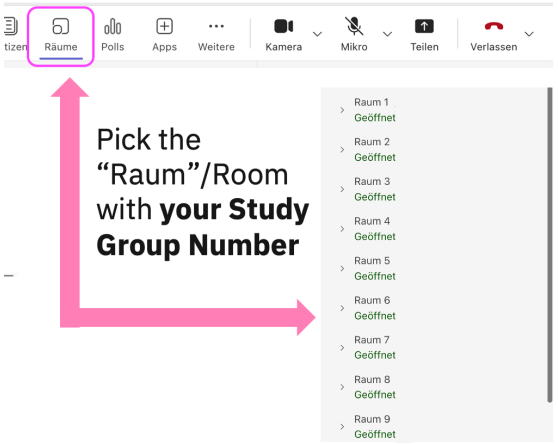


We'll meet **in the IBM Office** at Berlin Main Train Station (Ella-Trebe-Str. 3)



+

Online as always...you will have separate Teams rooms per study group or you join the main stage and see what we do in Berlin



Welcome & Agenda / Week 4

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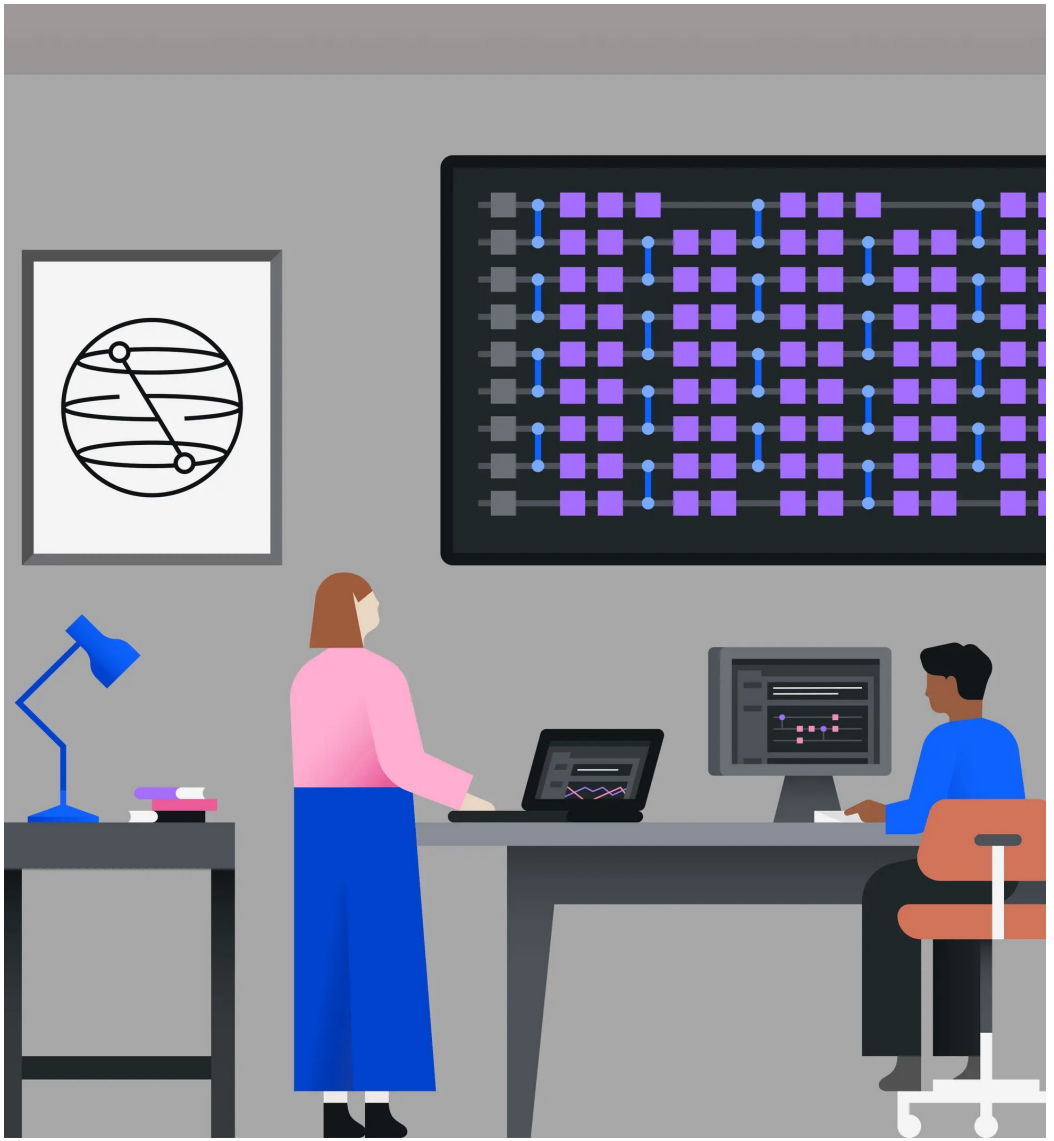
IBM Quantum Platform

Begin exploring quantum today with the Open Plan

Get up to 10 minutes of runtime per month free.

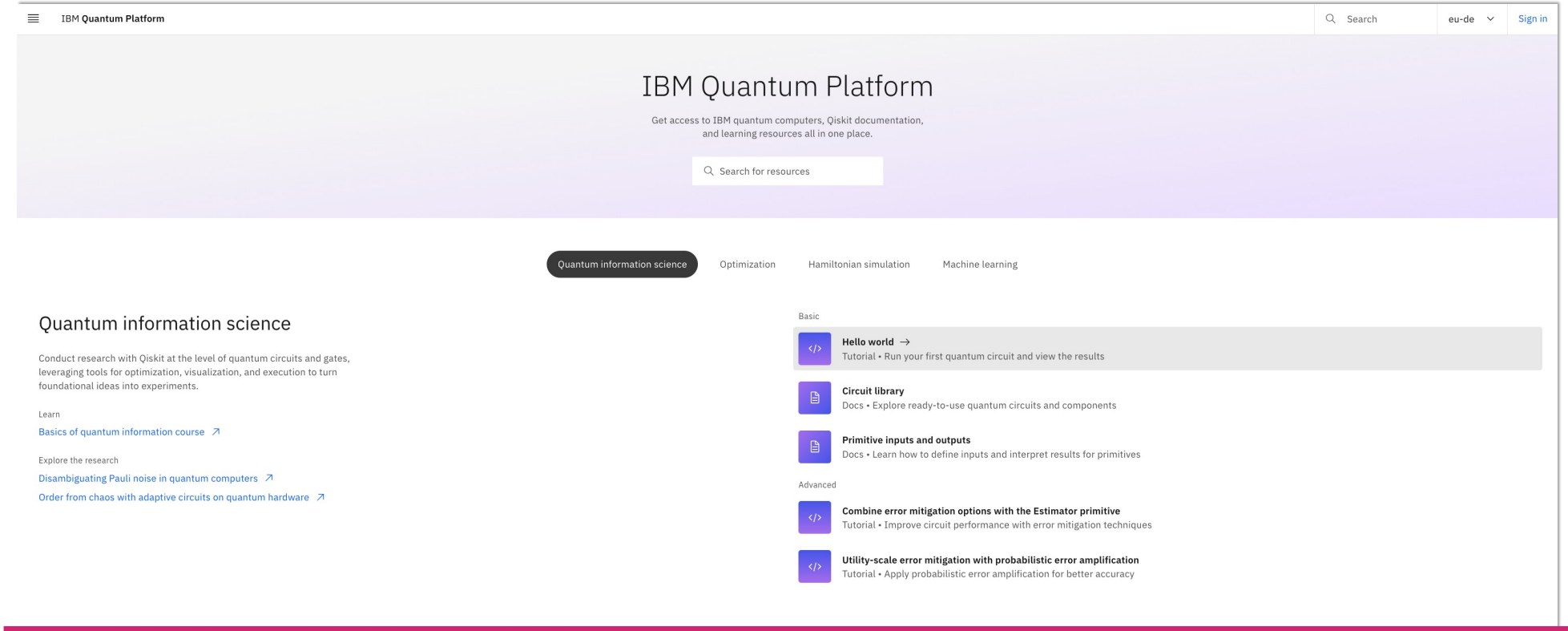
✓ Access to Qiskit Runtime as a service
✓ Access to 100+ qubit quantum computers

Feel free to create your free account on the IBM Quantum Platform now...



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IBM Quantum Platform.



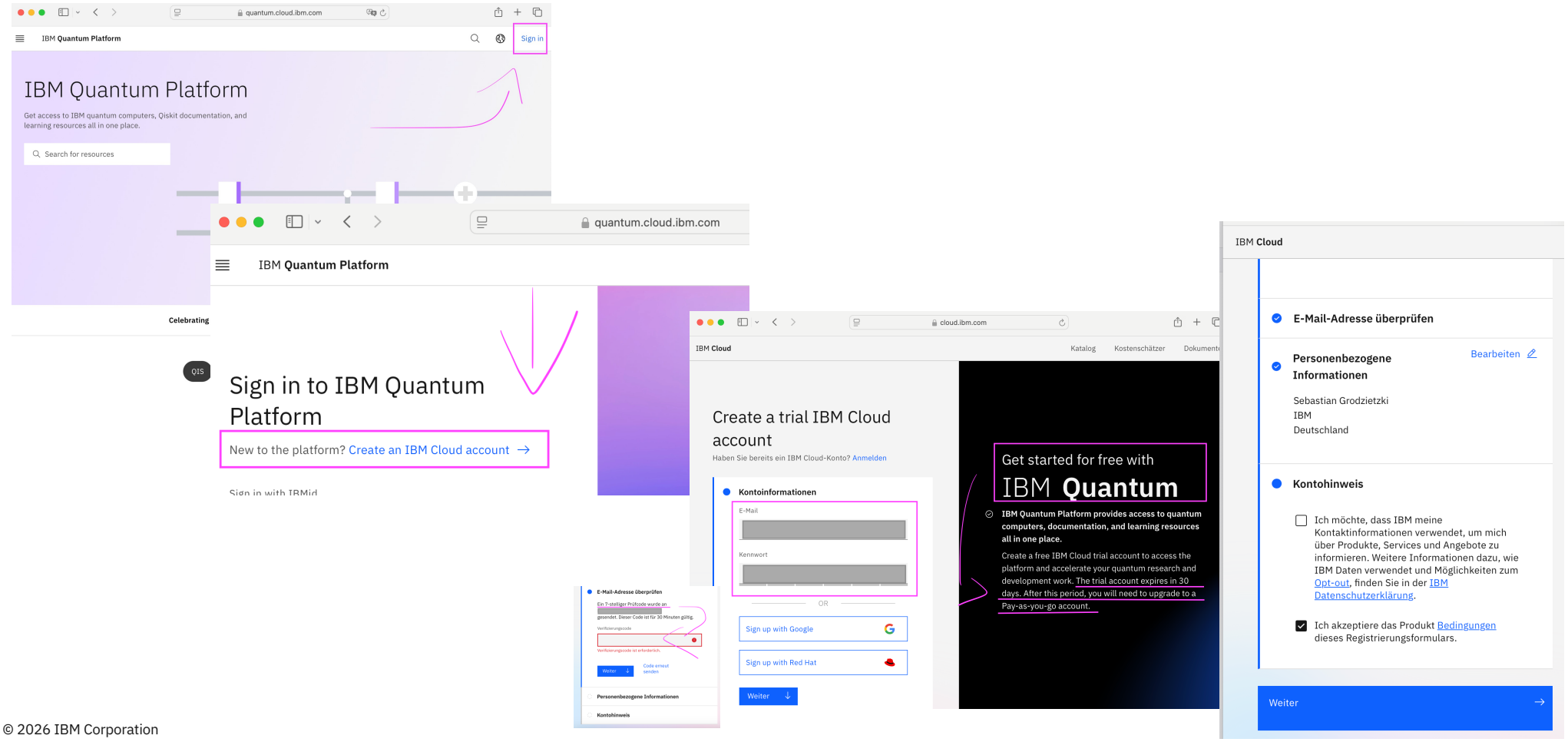
Documentation
Browse docs, APIs, and tutorials

Functions
Explore services to accelerate research

Composer
Graphically build and run circuits

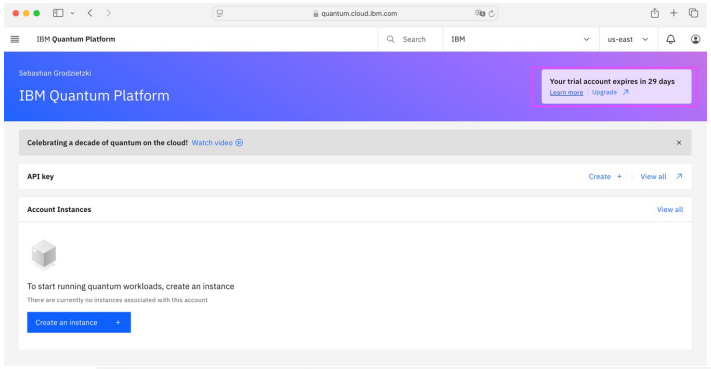
Announcements
View news, updates, and service alerts

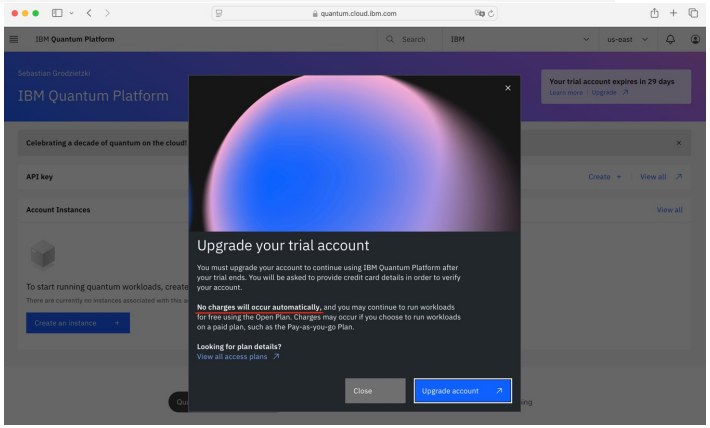
Open Plan Setup

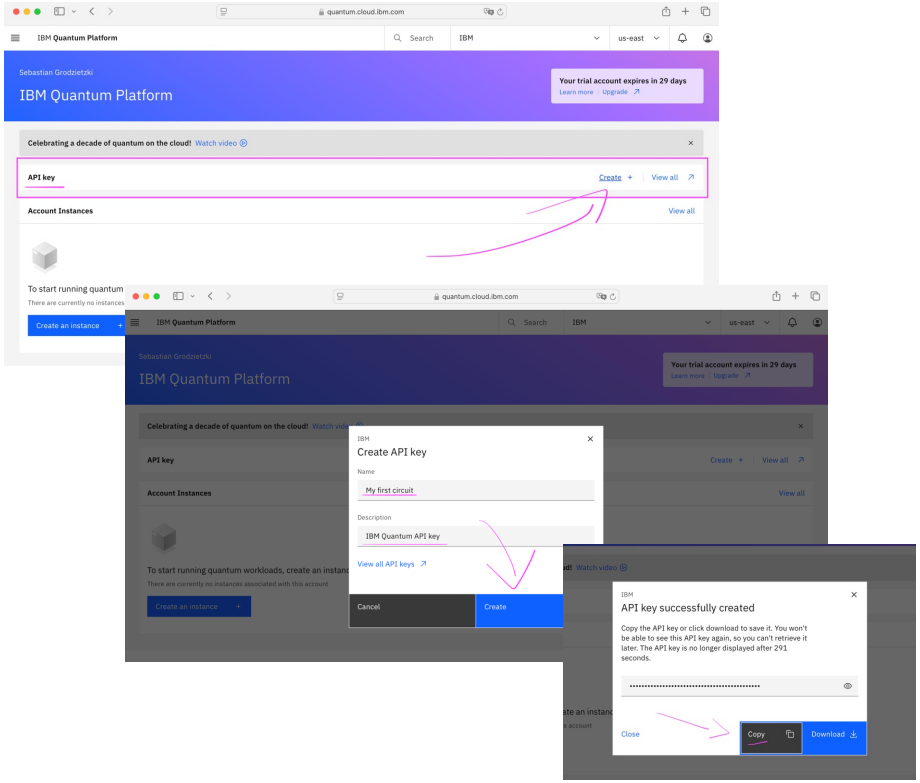


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Open Plan Setup

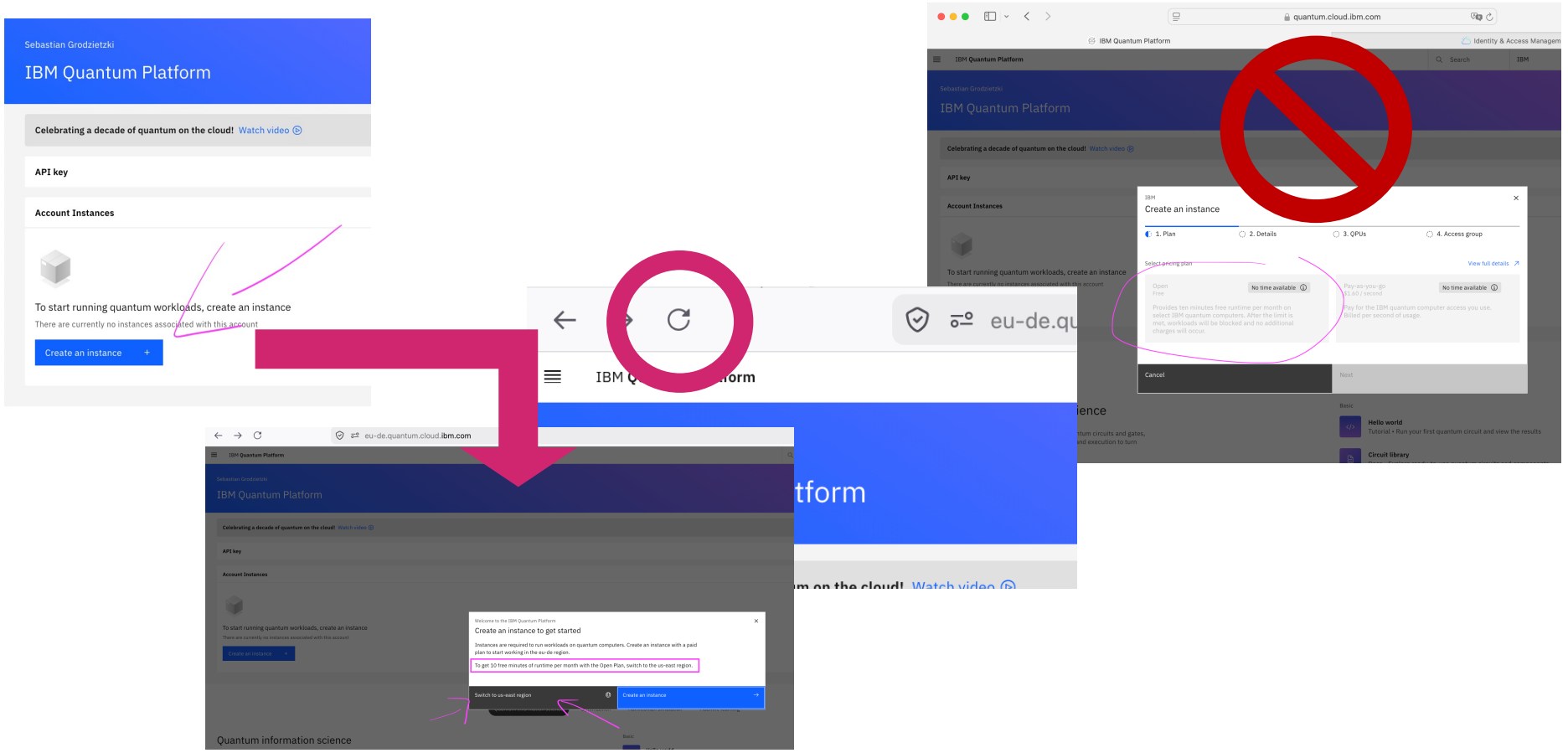






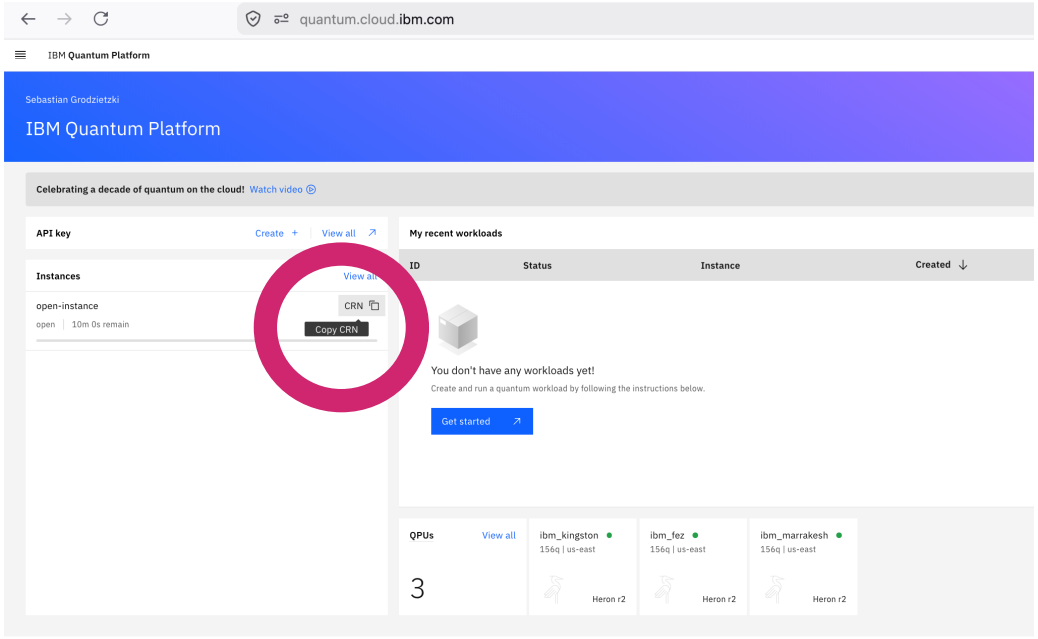
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Open Plan Setup



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Open Plan Setup



Your Open Plan Instance (CRN) and your API Token (your ID)

- Save your credentials locally with this code:

```
from qiskit_ibm_runtime import QiskitRuntimeService

QiskitRuntimeService.save_account(
    token="<your-api-key>", # Use the 44-character API_KEY you created and sa
    instance="<CRN>", # Optional
)
```

. Now you can use this Python code any time you want to authenticate to the Qiskit Runtime Service:

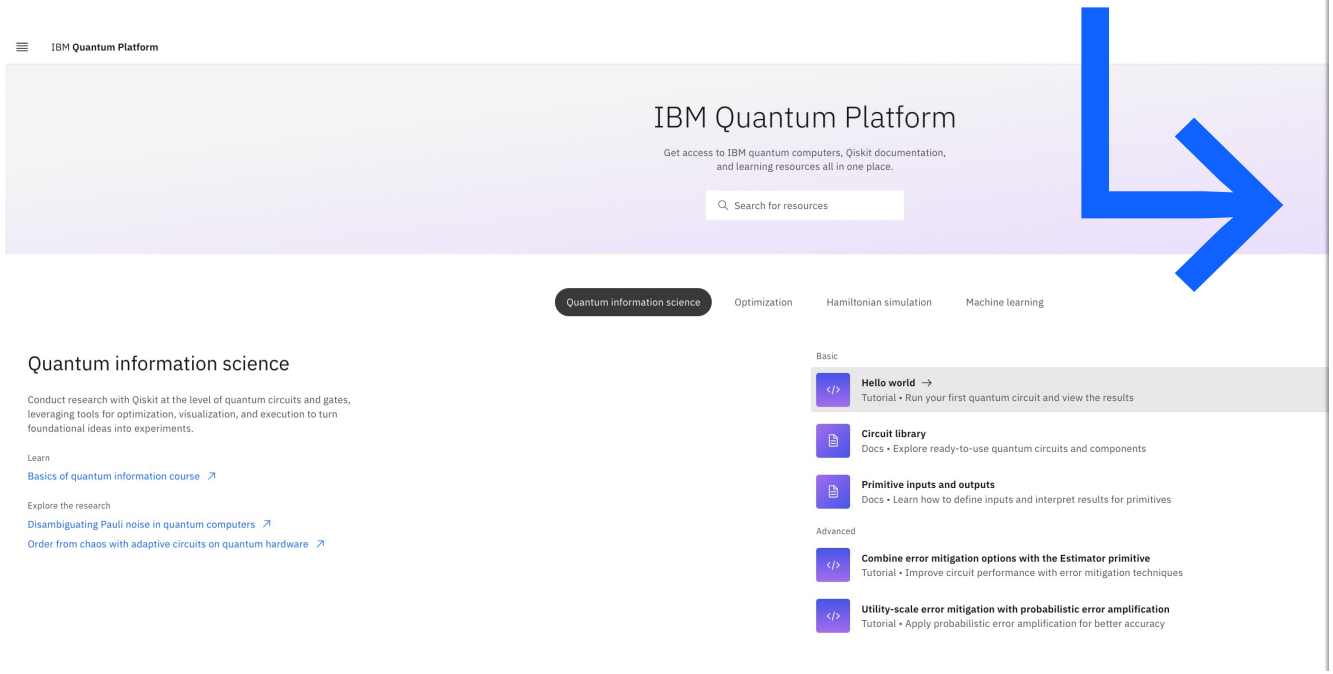
```
from qiskit_ibm_runtime import QiskitRuntimeService

# Run every time you need the service
service = QiskitRuntimeService()
```

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Let’s get started here:
IBM Quantum Platform.

https://quantum.cloud.ibm.com



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Coding Flavors

The practical path to useful quantum computing



Run quantum circuits accurately on quantum hardware

100+ qubit systems and software

Methods that provide execution of quantum circuits on hardware with higher accuracy



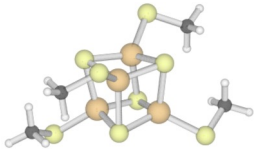
Map interesting problems to quantum circuits

Computational tasks that use those circuits

Industry and science applications that use those tasks

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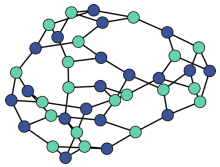
What are these problems?



Hamiltonian simulation

Simulating quantum mechanical processes found in nature to:

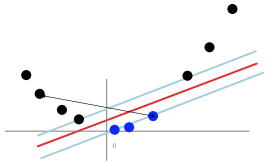
- Accelerate the discovery of new classes of antibiotics
- Develop lighter, longer-lasting batteries for electric vehicles, electronics, and energy grid storage



Optimization

Finding problem solutions within various constraints to:

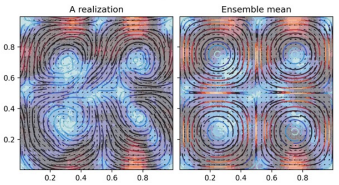
- Optimize portfolio construction to better manage risk
- Predict protein structures more accurately to design more targeted therapeutics



Machine learning

Improve machine learning performance with quantum techniques to:

- Enhance feature selection to optimize algorithmic trading
- Better understand physical and chemical properties of materials



Differential equations

New ways to solve differential equations difficult for classical to:

- Predict market performance with greater accuracy
- Simulate turbulence more accurately to improve aircraft and automotive design

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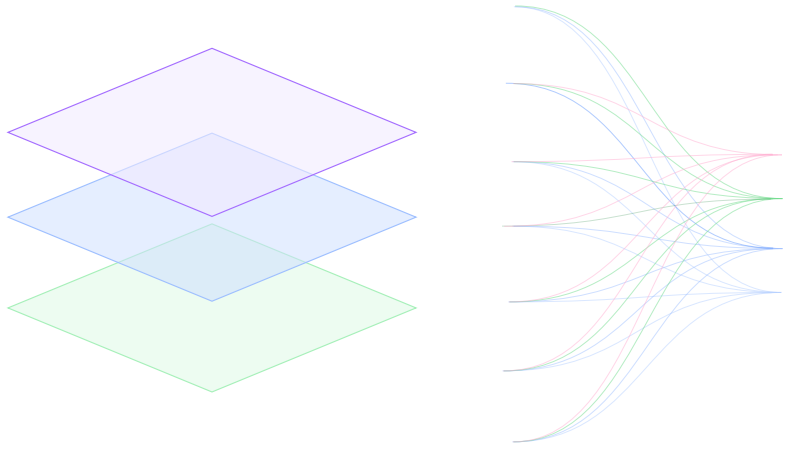
3 Problem Categories

↳ *3* ***totally*** *different Coding "****flavors****"*

Simulating Nature/ Chemistry & materials

Optimization

Quantum machine learning



↳ *but same start*

↳ *a best practice for coding all flavors*

↳ *and same outcome*

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The Start...

Start

A Hamiltonian / Cost Function: $\hat{H} = \hat{T}_e + \hat{T}_n + \hat{V}_{ee} + \hat{V}_{nn} + \hat{V}_{en}$

Total energy(H) of a quantum mechanical system = kinetic(T) + potential energy(V)

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The Start...

Start

A Hamiltonian / Cost Function: $\hat{H} = \hat{T}_e + \hat{T}_n + \hat{V}_{ee} + \hat{V}_{nn} + \hat{V}_{en}$

`Total energy(H) of a quantum mechanical system = kinetic(T) + potential energy(V)`

$\hat{T}_e$ **The kinetic energy of electrons**
This is pure energy of motion. Electrons are extremely light and whiz around the nuclei at tremendous speed.

$\hat{T}_n$ **The kinetic energy of the nuclei**
The motion of the heavy atomic nuclei (vibrations).

$\hat{V}_{ee}$ **Electron-electron repulsion (This is where Coulomb's law comes in!)**
This is Coulomb repulsion. Like charges (negative to negative) repel each other.

$\hat{V}_{nn}$ **Nuclear-nucleus repulsion**
The positive protons in the nucleus repel each other. Since the nuclei are firmly fixed in the crystal lattice, this is usually a constant energy value that stabilizes the system (like the framework of a house).

$\hat{V}_{en}$ **Nuclear-electron attraction**
This is the force that holds the atom together. The positive nuclei attract the negative electrons. Without this concept, the electrons would simply fly off into space.

The Start... Best Practice...

Start

A Hamiltonian / Cost Function: $\hat{H} = \hat{T}_e + \hat{T}_n + \hat{V}_{ee} + \hat{V}_{nn} + \hat{V}_{en}$

`Total energy(H) of a quantum mechanical system = kinetic(T) + potential energy(V)`

Best Practice

01. **Map** problem instance to quantum circuits and operators

Classical compute

02. **Optimize** for target hardware execution

Classical/Quantum Compute

03. **Execute** via Qiskit Runtime

Quantum Compute

04. **Result** processing / **Post-Process**

Classical Compute

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The Start... Best Practice...

Start

Best Practice

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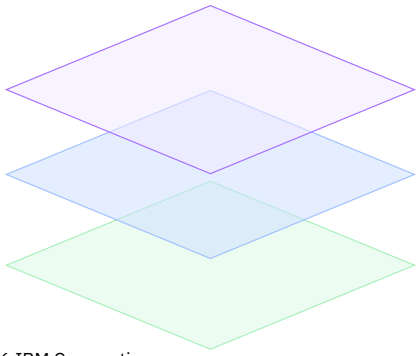
Classical/Quantum Compute

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Quantum Compute

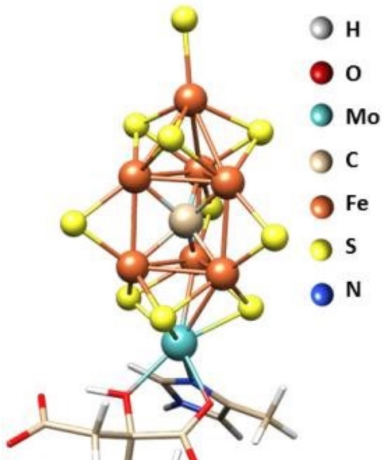
04. **Result** processing / **Post-Process**

Classical Compute



Simulating Nature:

„Import“ moleculs & atoms via Qiskit PySCF into a PCIDUMP file and let Qiskit map the electrons from the file to your Qubits to get the Hamiltonian and your Ansatz => The nature has been mapped for us to be our playground.



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The Start... Best Practice...

Start

Best Practice

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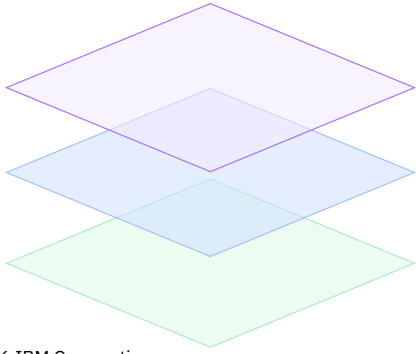
Classical/Quantum Compute

03. **Execute** via Qiskit Runtime

Quantum Compute

04. **Result** processing / **Post-Process**

Classical Compute



Optimization:

Design the map, the playground with all goals, conditions and „penalties" totally ourselves. E.g. take the TSP problem, where each route needs to be designed, the goal could be the shortest distance or least amount of trucks, etc. while visiting each house once (energy *penalties* if houses are being left out, if visited twice, if 2 trucks are on the same street, etc..)

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The Start... Best Practice...

Start

A Hamiltonian / Cost Function: $\hat{H} = \hat{T}_e + \hat{T}_n + \hat{V}_{ee} + \hat{V}_{nn} + \hat{V}_{en}$

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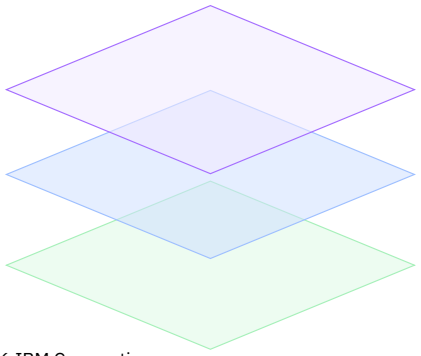
Classical/Quantum Compute

03. **Execute** via Qiskit Runtime

Quantum Compute

04. **Result** processing / **Post-Process**

Classical Compute



Quantum Machine Learning:
Encode your features (3 encoding techniques to save qubits) correlating with X, Y, Z qubit rotations to map them into the multidimensional Hilbert Space and then try to rotate the landscape so that we can draw a smooth plane (a dividing line) between "dog landscapes" and "cat landscapes".

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The Start... Best Practice... The Outcome...

Start

A Hamiltonian / Cost Function: $\hat{H} = \hat{T}_e + \hat{T}_n + \hat{V}_{ee} + \hat{V}_{nn} + \hat{V}_{en}$

`Total energy(H) of a quantum mechanical system = kinetic(T) + potential energy(V)`

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Classical compute

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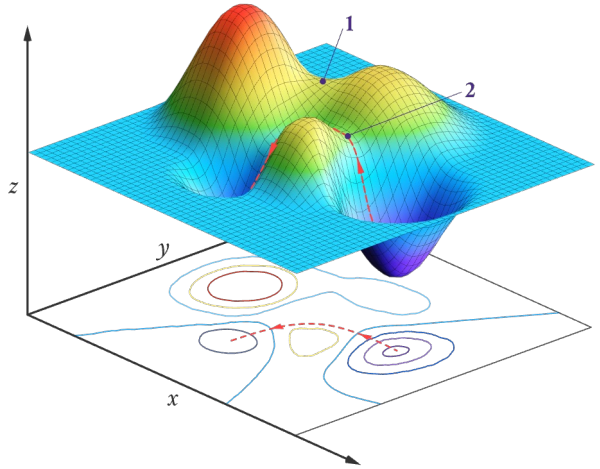
Quantum Compute

04. **Result** processing / **Post-Process**

Classical Compute

Outcome

An Energy Landscape



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Qiskit Coding Pattern & Tools

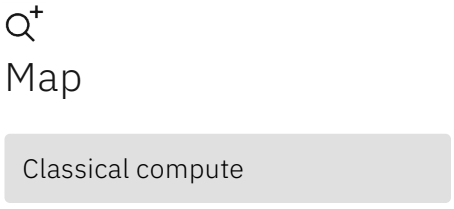
Qiskit Pattern:
The anatomy of a quantum algorithm

01
Map problem instance to quantum circuits and operators

02
Optimize for target hardware execution

03
Execute via Qiskit Runtime

04
Result processing



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Qiskit Coding Pattern & Tools

Qiskit Pattern:
The anatomy of a quantum algorithm

01
Map problem instance to quantum circuits and operators

Qiskit Circuit Library

Input:
Domain inputs

Output:
Circuits, observable

Map

Classical compute

02
Optimize for target hardware execution

Transpiler

Input:
Circuits, observable

Output:
ISA circuit, observable

Optimize

Classical/Quantum Compute

03
Execute via Qiskit Runtime

Primitives

Input:
ISA circuit, observable

Output:
Expectation value/samples

Execute

Quantum Compute

04
Result processing

Quantum Info

Input:
Expectation value/samples

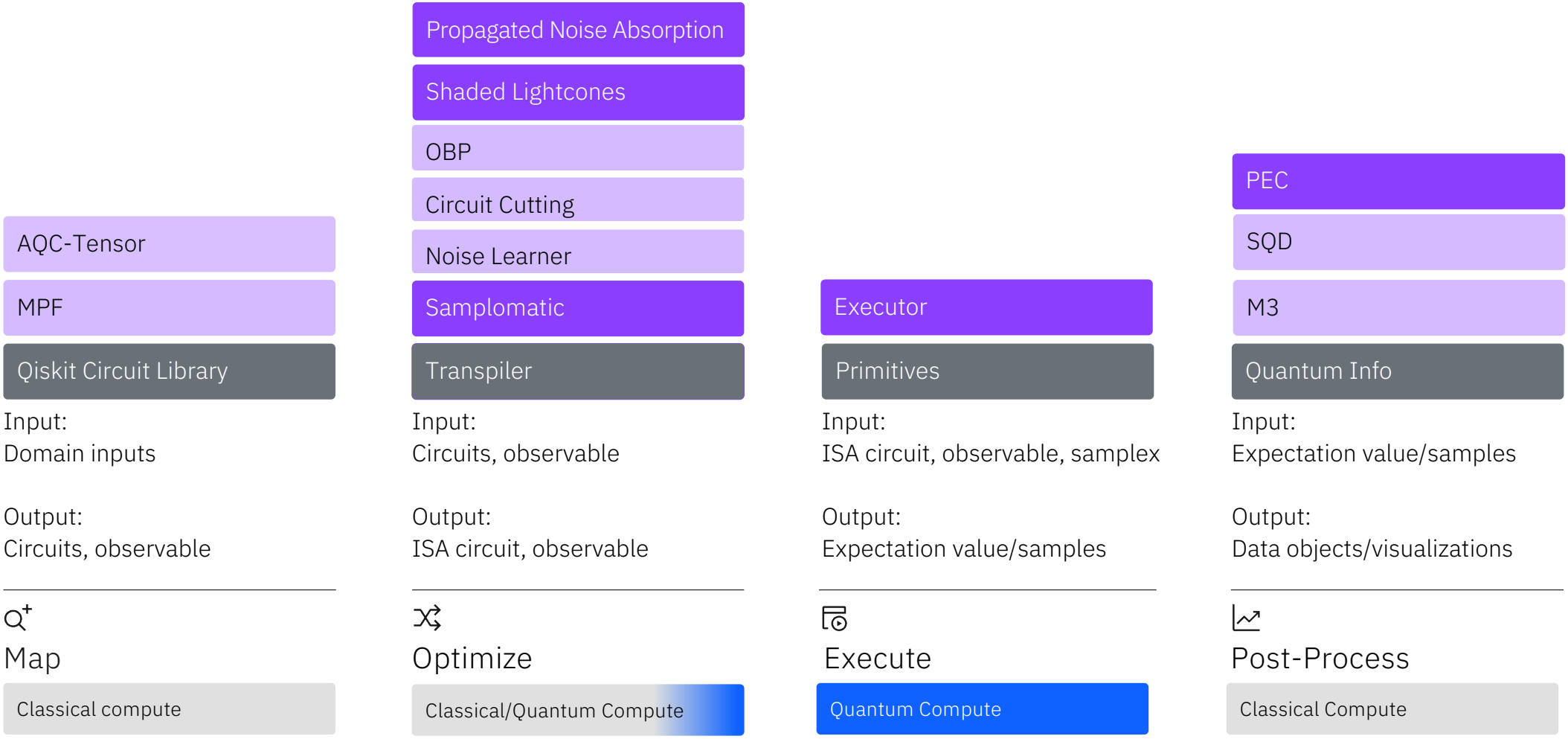
Output:
Data objects/visualizations

Post-Process

Classical Compute

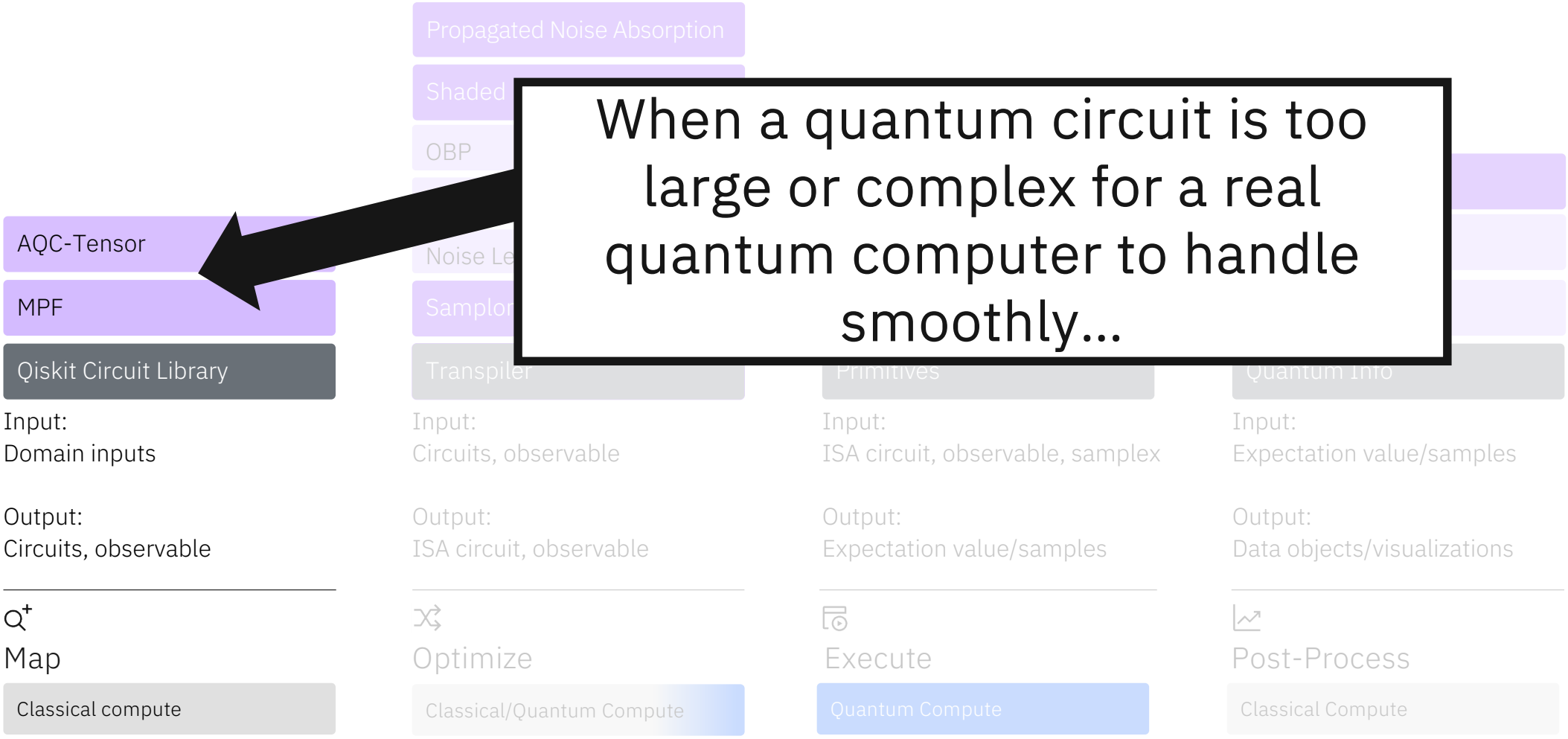
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Tools to mitigate errors / run complex circuits



© 2026 IBM Corporation

Tools to mitigate errors / run complex circuits



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Tools to mitigate errors / run complex circuits

Propagated Noise Absorption

Shaded Lightcones

OBP

Circuit Cutting

Noise Learner

Samplomatic

Transpiler

The circuit is rewritten and optimized so that the quantum hardware can understand it with the lowest possible error rate.

AQC-Tensor

MPF

Qiskit Circuit Library

Input:
Domain inputs

Output:
Circuits, observable

Map

Classical compute

Input:
Circuits, observable

Output:
ISA circuit, observable

Optimize

Classical/Quantum Compute

Input:
ISA circuit, observable, samplex

Output:
Expectation value/samples

Execute

Quantum Compute

Input:
Expectation value/samples

Output:
Data objects/visualizations

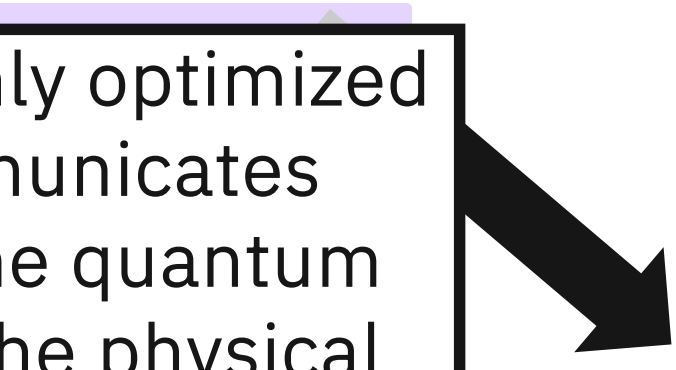
Post-Process

Classical Compute

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Tools to mitigate errors / run complex circuits

It takes the highly optimized circuit, communicates directly with the quantum chip, triggers the physical pulses, and ensures the hardware executes the instructions perfectly.



Output:
Circuits, observable

Map

Classical compute

Output:
ISA circuit, observable

Optimize

Classical/Quantum Compute

Executor

Primitives

Input:
ISA circuit, observable, samplex

Output:
Expectation value/samples

Execute

Quantum Compute

PEC

SQD

M3

Quantum Info

Input:
Expectation value/samples

Output:
Data objects/visualizations

Post-Process

Classical Compute

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Tools to mitigate errors / run complex circuits

The quantum computer has finished its job, but the raw data is still "dirty" and full of noise. These tools filter out the errors.



PEC

SQD

M3

Quantum Info

Input:
Expectation value/samples

Output:
Data objects/visualizations

Post-Process

Classical Compute

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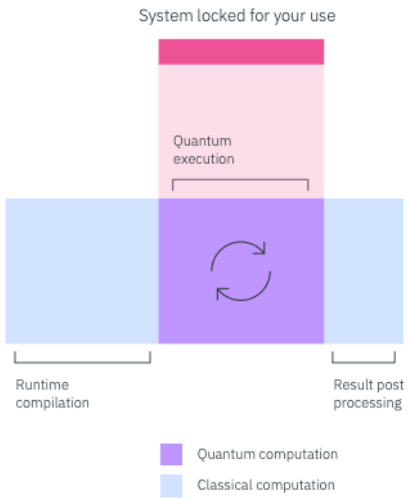
Modes of circuit execution

Job

A single primitive request that contains all the context for executing your workload

Example use case:

Verification of circuits

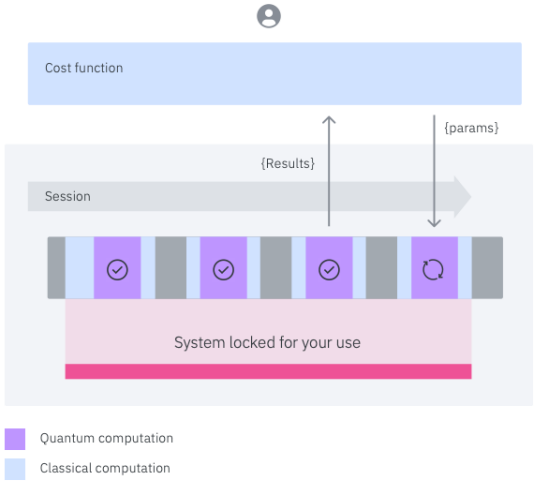


Session

A multi-job workload that uses results of last job to conditionally update inputs of subsequent jobs until it converges

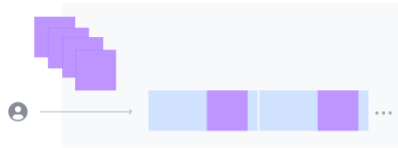
Example use case:

VQE, QAOA

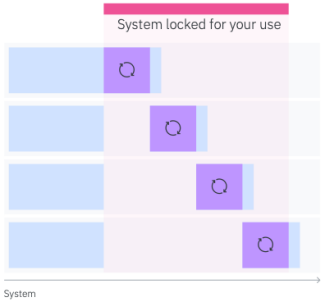


Batch

Users can now batch workloads and the service will efficiently thread jobs to reducing time to results.



```
with Batch(backend = backend) as batch:
        estimator = Estimator(batch)
        job1 = estimator.run(qc, obs, params1)
        job2 = estimator.run(qc, obs, params2)
       # …
        jobN = estimator.run(qc, obs,paramsN)
```



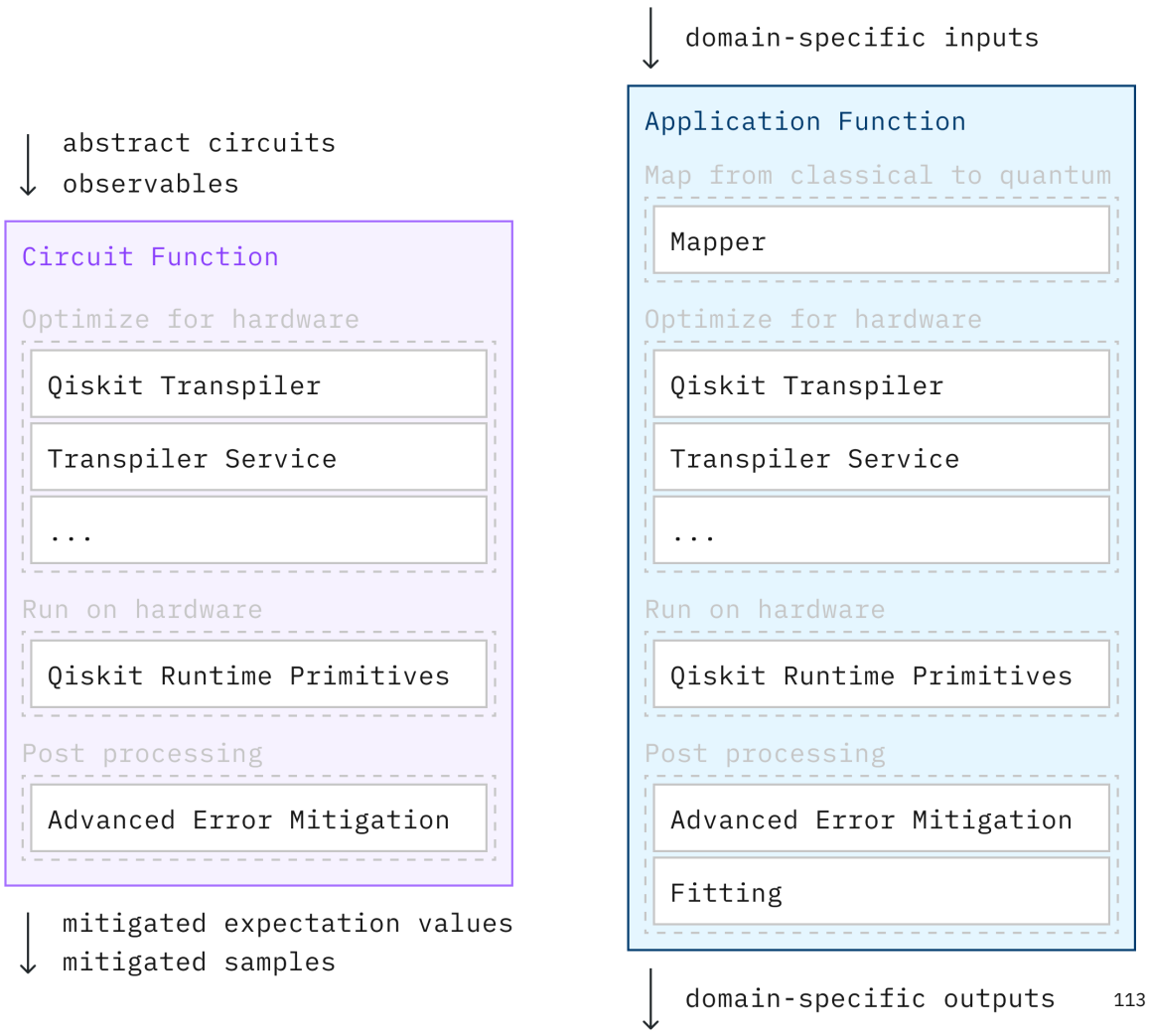
Quantum computation
Classical computation

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Qiskit Functions are abstracted services, designed to accelerate development

IBM is partnering with the broader quantum computing ecosystem to offer a catalog of managed, utility-scale services to unlock new possibilities:

- Circuit Functions: Enabling quantum computational scientists to discover new algorithms and applications, without needing to manage transpilation, error suppression, or error mitigation
- Application Functions: Enabling data scientists and enterprise developers to integrate quantum into industry workflows, while leveraging familiar domain abstractions



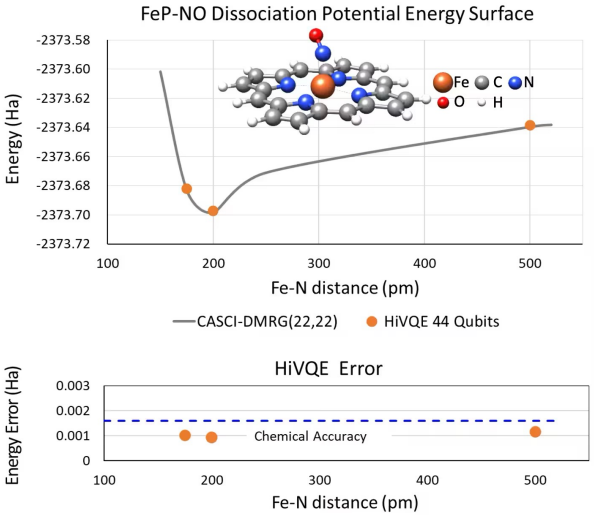
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Exploring applications with **Qiskit Functions Catalog**

Simulation

QUNOVA COMPUTING

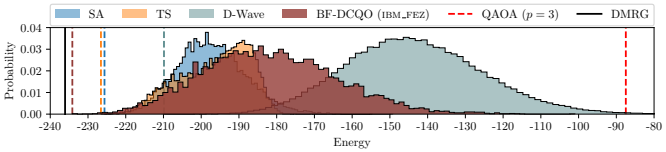
Potential energy surface (22o, 22e), with chemical accuracy (> 1.6 mHa).



Optimization

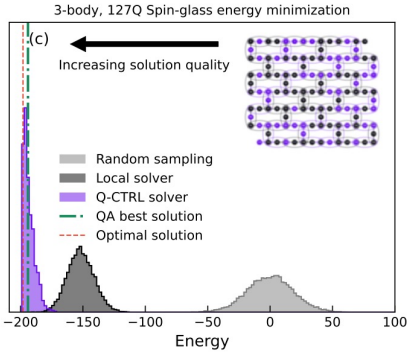
KIPU QUANTUM

Higher-order unconstrained binary optimization closest to ideal answer, beating D-wave, QAOA, and several classical methods



Q-CTRL

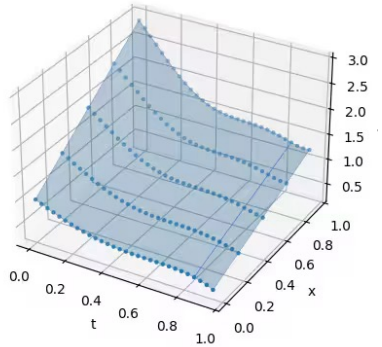
1500x more likely to find ground-state solution, compared to annealers



Differential equations

ColibriTD

Computational Fluid Dynamics at 50Q, with linear growth of computational resources



Machine learning

MULTIVERSE COMPUTING

Financial risk management and other classification problems at 100Q

Finance

Global Data Quantum

Dynamic portfolio optimization of mixed assets at 100Q+ scale

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Qiskit Advocates Program

A global cohort of 1300+ of the strongest Qiskit Community members who receive support, exclusive opportunities, mentorship, and much more from IBM Quantum

Mentorship

Advocates create long-lasting relationships with IBMers and industry experts via long-term collaborative mentorship projects, contributed to the open source community

Networking

Advocates are invited to join a group of quantum experts, who routinely share opportunities and updates from around the world

Events

Advocates travel to conferences, present at community engagements, host their own events, and mentor others through hackathons and quantum challenges

Recognition

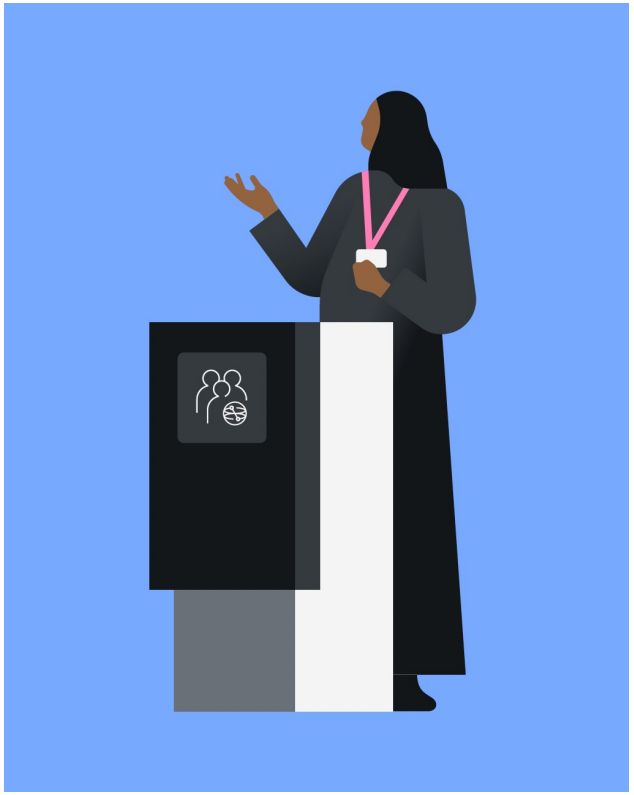
Advocates are seen by many as the strongest members of the field, thanks to the Qiskit Developer Certification, community contributions, and success of the group at large

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What is the Qiskit advocate program?



Join 1300+ Qiskit Advocates across 50 countries in advancing quantum computing education and community.



The Qiskit advocate program is an external, global initiative that empowers and enables aspiring leaders in the Qiskit open-source community.

Qiskit advocates enhance their professional growth by:

- Building meaningful connections with quantum experts and peers
- Gaining exclusive access to Qiskit education and events
- Getting unique opportunities to contribute to quantum computing advancement
- Making open-source contributions and sharing knowledge through mentorship
- Participating in Qiskit events, networking, and quantum research and so much more.

Perfect for: Students, software developers, quantum researchers, educators, and anyone looking to upskill and advance their career in quantum computing.

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Qiskit Global Summer School 2026

13–24 July 2026

annual virtual replay

© 2026 IBM Corporation

Qiskit Global Summer School 2026: Registration now open

Sign up for **Qiskit Global Summer School 2026: A decade on the cloud**—a free, virtual program for learning quantum computing with Qiskit.



Authors
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Olivia Lanes
Marcel Pfaffhauser
Robert Davis

Topics
Community Qiskit

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Registration is now open for **Qiskit Global Summer School 2026: A decade on the cloud**—the latest entry in an annual, global program designed to help students, researchers, and developers build practical skills in quantum computing. The two-week event will run from **13–24 July 2026.**

Register for Qiskit Global Summer School 2026

It's been ten years since IBM put the first quantum computer on the cloud, giving rise to a global ecosystem of developers, researchers, startups, and enterprises all working to advance quantum technology. For this year's Qiskit Global Summer

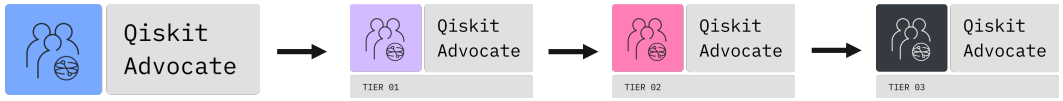
Key takeaways:

- Register now for Qiskit Global Su School 2026: A decade on the clo July 2026).
- The free, online program builds p quantum skills with Qiskit throug labs, live Q&As with IBM Quantur and more.
- Flexible, fully virtual format inclu recorded content, interactive labs global Discord community.
- New features include a beginners updated scoring requirements, ar

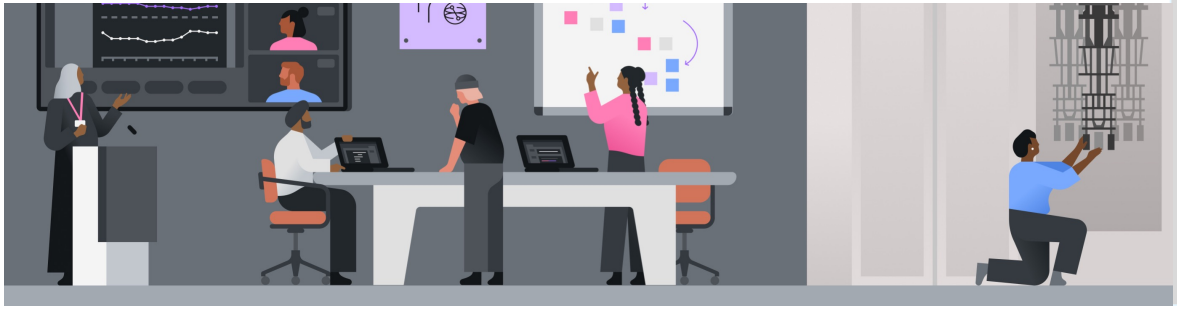
What do Qiskit advocates do?

Activities:

- Give talks and presentations on Qiskit at events and conferences
- Contribute to the Qiskit SDK and Qiskit Ecosystem projects
- Mentor participants in flagship events and host events in your local community
- Provide feedback on Qiskit and IBM Quantum tools as active users
- Collaborate with advocates globally
- Earn points through community contributions to advance through program tiers



For more information, see:
ibm.biz/advocates-application



Ready to get started?

Applications are now open:

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Q&A

Thank you!!

See you next Wednesday, for the first **flipped classroom** *in the IBM office* Berlin *and online,* same place, 13:00-14:30

Sebastian E. Grodzietzki, Dipl.-Inf.
Joey Rogers, M.Sc. Physics